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Up in the Heavens and Down in the Earth

EXPERIMENTING WITH SPACE AND TIME

We've come a long way since Empedocles of Agrigento explained the universe using earth, air, fire, and water. And much of the progress we've made, from Newton through the revolutionary discoveries of the twentieth century, has been borne out spectacularly by experimental confirmation of detailed and precise theoretical predictions. But since the mid-1980s, we've been the victims of our own success. With the incessant urge to push the limits of understanding ever further, our theories have entered realms beyond the reach of our current technology.

Nevertheless, with diligence and luck, many forefront ideas will be tested during the next few decades. As we'll discuss in this chapter, experiments either planned or under way have the potential to give much insight into the existence of extra dimensions, the composition of dark matter and dark energy, the origin of mass and the Higgs ocean, aspects of early-universe cosmology, the relevance of supersymmetry, and, possibly, the veracity of string theory itself. And so, with a fair bit more luck, some imaginative and innovative ideas regarding unification, the nature of space and time, and our cosmic origins may finally be tested.

Einstein in Drag

In his decade-long struggle to formulate the general theory of relativity, Einstein sought inspiration from a variety of sources. Most influential of all were insights into the mathematics of curved shapes developed in the nineteenth century by mathematical luminaries including Carl Friedrich Gauss, János Bolyai, Nikolai Lobachevsky, and Georg Bernhard Riemann. As we discussed in Chapter 3, Einstein was also inspired by the ideas of Ernst Mach. Remember that Mach advocated a relational